~~Obs:~~ The blocks inside ~~B_G~~ B_G^{**} are minimally covered and robust.

The set of ~~all~~ regionally robust blocks is setwise invariant under all automorphisms

A l.c.s.c. group has at most $\aleph_0$ regionally robust ~~blo~~ blocks.

A l.c.s.c. block has at least one re. ro block provided that its structure as a top group is sufficiently complex.

Prop:

For G a l.c.s.c. group, either

(i) $B_G^{**} \neq \emptyset$, or

(ii) G° is compact-by-solvable-by-compact, and G/G° is elementary with rank at most ~~ω~~ $\omega+1$. If G is compactly generated, then G/G° is elementary with finite rank.

AA

Thrm:

Suppose that G is a Topologically characteristically simple l.c.s.c. groups. Then G is exactly one of the following:

(i) an abelian group;

(ii) a non-abelian quasi-discrete t.d.l.c s.c. group;

(iii) an elementary group of decomposition ~~with~~ rank ~~2~~ 2 with trivial quasi-center;

(iv) an elementary group ~~with rank~~ of decomposition rank $\omega+1$;

(v) a direct product of copies of simple Lie groups;

(vi) a quasi-product of copies of a topologically simple t.d.l.c.s.c group of decomposition rank greater than $\omega+1$;

(vii) a t.d.l.c.s.c. group of stacking type and decomposition rank greater than $\omega+1$.

In particular, chief factors of l.c.s.c groups has one of the above forms.

2.1 / Notations and definitions

Def (Directed family)

For $\mathcal{P}$ a poset, we say $\mathcal{D} \subseteq \mathcal{P}$ is a directed family if $\forall M, N \in \mathcal{D}$ there is $L \in \mathcal{D}$ with $M \leq L$ and $N \leq L$. We say that $\mathcal{F} \subseteq \mathcal{P}$ is a filtering family if for all $M, N \in \mathcal{F}$ there is $L \in \mathcal{F}$ s.t. $L \leq M \cap N$.

Notations

Given a group G and a set I, we denote the direct sum by $G^{<I}$ and the direct product by G^{I}. For a subset $K \subseteq G$, $C_G(K)$ is the collection of elements of G that centralizes every element of K. For the normalizer we write $N_G(K)$. For $A, B \subseteq G$, we put

$$[A, B] := \langle aba^{-1}b^{-1} \mid a \in A, b \in B \rangle.$$

For $a, b \in G$, $[a, b] := aba^{-1}b^{-1}$, and $a^b = bab^{-1}$. For $a, b, c \in G$ we put $[a, b, c] = [[a, b], c]$.

All top groups are Hausdorff etc. etc.

G° is the connected component of ~~*~~ 1

Def (Perfect)

We will say ~~&~~ G is top perfect if $[G,G]$ is dense in G.

Def (Top. characteristically simple)

If the only proper subgroup of G that is preserved by all ~~automorphs~~ automorphisms of G is the trivial one.

Def (Polish)

If separable via the metric that induces the ~~t~~topology.

2.2 / Generalities on l. c. groups

dfn: (Almost connected)

If G° is cocompact.

For G a l.c. group, the set of almost connected open subgroups is denoted by $\mathcal{U}(G)$

↳ obs: ~~∀~~ $U \leq G$ is almost connected iff $U/G^\circ \cap U$ is comp.

If G is a t.d. l.c. group, then $\mathcal{U}(G)$ is a base of identity neigh.

dfn: (Locally elliptic)

Every finite subset generates compact subgroup.

$$Rad_{LE}(G) = \bigcup \{ H \trianglelefteq G ; \ H \text{ is elliptic} \}$$

Thrm:

For G a l.c. group, $Rad_{LE}(G)$ is the unique largest locally elliptic closed normal subgroup of G. Additionally

$$Rad_{LE}(G / Rad_{LE}(G)) = \{1\}$$

Def: (Quasi-center)

$$QZ(G) = \{g \in G ; C_G(g) \text{ is open}\}.$$

We also say that a l.c. group is quasi-discrete if it has a dense quasi-center.

Prop:

Let G be a compactly generated t.d.l.c. group, $U \in \mathcal{U}(G)$, let A be a compact symmetric set of G s.t. $G = \langle U, A \rangle$ and let D be a dense symmetric subset of G.

(i) $\exists B \subseteq_{\text{finite}} G$ finite symmetric s.t. $B \subseteq D$ and

$$BU = UB = UBU = UAU$$

(ii) for any subset B satisfying (i) it is the case that $G = \langle B \rangle U$ and the coset space G/U carries the structure of a locally finite connected graph, invariant under the natural G-action, where gH is adjacent to hH iff $gH \neq hH \wedge Ug^{-1}hH \subseteq UBU$. Thus, G/N is the vertex set of a Cayley-Abels graph for G.

Obs: A Lie group L has a ~~maximal~~ largest connected ~~component~~ solvable normal subgroup; it is called solvable radical of L and denoted by $\mathrm{Sol}(L)$.

A connected lie group with trivial solvable radical is called semisimple. A semisimple group has discrete center, and modulo its center, it is a finite direct product of abstractly simple groups.

The manifold structure of a Lie group gives a well-defined dimension $\dim_{\mathbb{R}}(G)$, which is additive with respect to extensions. If $\dim_{\mathbb{R}}(G) = 0 \Rightarrow G$ is discrete. The dimension places a restriction on ~~normal~~ directed families of normal subgroups.

Lemma:

Let G be a connected lie group and let $\mathcal{R}$ be a directed family of closed normal subgroups of G. Then there exists $N \in \mathcal{R}$ s.t. MN/N is discrete in G/N for all $M \in \mathcal{R}$.

↳ Proof:

Let $d = \sup\limits_{N \in \mathcal{D}}(\dim_{\mathbb{R}}(N))$ and take $N \in \mathcal{D}$ realizing this maximum. For all $M \in \mathcal{D}$, there is $K \in \mathcal{D}$ extending N and M, since $\mathcal{D}$ is directed. The group K/N is a dimension zero lie subgroup of G/N, hence K/N is discrete. We thus conclude that MN/N is also discrete, verifying the lemma. ∎

Thrm:

If G is an almost connected l-c- group, then $\mathrm{Rad}_{\mathcal{LE}}(G)$ is compact, and $G/\mathrm{Rad}_{\mathcal{LE}}(G)$ is a Lie group. Moreover, there exists arbitrarily small compact normal subgroups of K of G s.t. G/K is a Lie group.

2.3/ Normal compression of Polish groups

Defn: (Normal compression map)

For G and H Polish groups, a normal compression map from G to H is an inj. cont. hom $\psi: G \hookrightarrow H$ with dense, normal image. When such a map exists, we say H is a normal compression of G.

Obs: Given a normal compression map $\psi: G \to H$, there is a canonical action $H \curvearrowright G$ via

$$h.g := \psi^{-1}(h\,\psi(g)\,h^{-1}).$$

This action is called ψ-equivariant action, as it is with respect to which ψ is an H-equivariant map for H acting on itself by conjugation.

Prop:

Suppose G and H are Polish groups with $\psi: G \to H$ a normal compression map and let $O \leq_o H$. Then the following hold:

(i) $\pi: G \rtimes_\psi O \longrightarrow H$ via $(g, o) \longmapsto \psi(g)o$ is a cont. surj. homom with $\mathrm{Ker}(\pi) = \{(g^{-1}, \psi(g)) \mid g \in \psi^{-1}(O)\}$ and if

$Q = H$, then $\mathrm{Ker}(\pi) \simeq G$ as top groups;

(ii) $\psi = \pi \circ i$ where $i : G \hookrightarrow G \rtimes_\psi Q$ is the usual inclusion;

(iii) $G \rtimes_\psi Q = \overline{i(G)\,\mathrm{Ker}(\pi)}$, and the subgroups $i(G)$ and $\mathrm{Ker}(\pi)$ are closed normal subgroups of $G \rtimes_\psi Q$ with trivial intersection.

$$\begin{tikzcd}
\{1\} \arrow[r] & G \arrow[r, hook] \arrow[rr, dashed, bend right, "\psi"'] & G \rtimes_\psi Q \arrow[r, two heads] & H \arrow[r] & 1
\end{tikzcd}$$

Prop: Suppose that G and H are Polish groups with $\psi : G \to H$ a normal compression map and let K be a closed normal subgrp. of G. Then the following hold:

(i) The image $\psi(K)$ is a normal subgroup of ~~G~~ H;

(ii) If $\psi(K)$ is also dense in H, then $\overline{[G,G]} \leq K$, and every closed normal subgrp of K is normal in G.

Lemma: Let G and H be Polish groups with $\psi : G \to H$ a normal compression map. Let $R := \psi^{-1}(H^\circ)$ and let $N := \overline{\psi(R)}$. Then the following inclusion holds:

$$[H^\circ, \psi(G)] \leq \psi(R); \; [H^\circ, H] \leq N; \; \overline{[H^\circ, \psi(G)]} \leq \psi(G^\circ); \; [H^\circ, N] \leq \overline{\psi(G^\circ)}.$$

Thm: Suppose that G and H are Polish groups with $\psi: G \to H$ a normal compression map. Then the following hold:

(i) The group $H^\circ / \overline{\psi(G^\circ)}$ is nilpotent of class at most two;

(ii) if H is connected, then both G/G° and $H/\overline{\psi(G^\circ)}$ are abelian;

(iii) if G is t.d., then H° is central in H.

Coro: Suppose G and H are centerless topologically perfect Polish groups and that H is a normal compression of G. Then G is connected iff H is connected, and G is t.d. if H is t.d.

Elementary grps and decomposition rank

3.1/ Preliminaries

Elementary grps are etc.

__Lemma:__ Let G be comp. gen. t.d.l.c. group s.t. $Res(G)=\{1\}$. Then G admits a base of neigh of the identity consisting of compact-open normal subgroups. It follows that for any closed normal subgroup K of G the quotient G/K has a base of neigh. of the identity consisting of compact open normal subgroups and in particular $Res(G/K)=\{1\}$.

3.2/ Stability of the decomposition rank

__Lemma:__ Suppose $H_1, \dots, H_n$ are t.d l.c. s.c. groups. If each H_i is elementary, then $\prod_{i=1}^{n} H_i$ is elementary with

$$\xi(H) = \max_{1 \leq i \leq n} \xi(H_i). \qquad \left(H = \prod_{i=1}^{n} H_i \right)$$

↳ __Proof:__ Because ~~...~~ $H_i \trianglelefteq H$ we get $\max_i \xi(H_i) \leq \xi(H)$

We will prove by induction the other side. If ~~...~~ $\xi(H)=1$ then H is the trivial group.

Suppose $\xi(H) = \beta + 1$. For each ~~...~~ $1 \leq i \leq n$ let $(O_{j,i})_{j\in\mathbb{N}}$ be an increasing exhaustion of H_i by compactly generated open subgroups. The groups $L_j := \prod_{i=1}^{n} O_{i,j}$ then form an increasing exhaustion sequence of G by compactly generated open subgrps. Hence, $\mathcal{G}(H) = \sup_{j \in \mathbb{N}} \mathcal{G}(Res(L_j)) + 1$.

On the other hand, $Res(L_j) = \prod_{i=1}^{n} Res(O_{i,j})$, so the inductive hypothesis ensures that

$$\mathcal{G}(Res(L_j)) = \max_{1 \leq i \leq n} \mathcal{G}(Res(O_{i,j})).$$

By passing to a subsequence if necessary, we may assume there is $1 \leq K \leq n$ s.t. $\mathcal{G}(Res(L_j)) = \mathcal{G}(Res(O_{j,K}))$ for all $j \in \mathbb{N}$. Therefore

$$\mathcal{G}(H) = \sup_{j \in \mathbb{N}} \mathcal{G}(Res(O_{j,K})) + 1 = \mathcal{G}(H_K$$

Since it must be the case that $\mathcal{G}(H_K) = \max_i \mathcal{G}(H_i)$, the induction is complete. ∎